When Does Representation Geometry Predict Distance-Based OOD Ranking? An Invariance-Aware Intervention Study

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Abstract

Distance-based out-of-distribution (OOD) detection is often interpreted through global properties of a learned representation, including intrinsic dimensionality and covariance-spectrum decay. Evidence directly connecting these properties to OOD ranking is commonly observational: representations compared across backbones differ simultaneously in architecture, pretraining data, objective, scale, and normalization. Moreover, a geometric change can be irrelevant to a scorer when it lies inside that scorer's invariance class, and a near-ceiling baseline can leave too little ranking headroom for an effect to be observed. We study these issues through a controlled intervention on frozen PanDerm representations, using ISIC 2019 as the in-distribution population and PAD-UFES-20 as an exposed cross-modality stress-test domain. We formalize scorer-relative rank invariance, derive quantitative bounds for ridge-induced Mahalanobis violation and anisotropic cosine distortion, and connect bounded score perturbations to AUROC changes through local ID–OOD ranking margins. A gated dose ladder then separates manipulation and utility admissibility from scorer visibility and empirical susceptibility. Spectral slope moves reproducibly and nearly monotonically, while LID changes reproducibly but non-monotonically and in-distribution task utility remains above the prespecified threshold. Neither target metric has a Holm-corrected configuration-level association with scorer AUROC. Cosine and KDE are descriptively closer to intervention-history-matched than slope-matched configurations, but no paired contrast survives correction. The contribution is therefore an admissibility framework rather than a universal null claim.

Keywords: Out-of-distribution detection; representation geometry; scorer invariance; Mahalanobis distance; nearest neighbors; intervention study; foundation models.

1 Introduction

Out-of-distribution detection is a safeguard for machine-learning systems deployed beyond their development population. For feature-space detectors, operational behavior depends on an interface between the representation produced by the encoder and the scoring functional that reads it. A representation can therefore appear geometrically favorable under a global summary while remaining poorly aligned with the ranking rule used for detection.

Distance-based scorers make this interface explicit. Mahalanobis scoring reads displacement through an estimated precision matrix [1]; cosine-to-centroid scoring retains angles while discarding magnitude; nearest-neighbor scoring depends on local order statistics [2]; and kernel-density

estimation additionally depends on bandwidth and local mass. These scorers do not respond to transformations in the same way. A shared invertible affine map can substantially change Euclidean spectra while leaving an unregularized, consistently refitted Mahalanobis distance unchanged. Conversely, anisotropic scaling can alter angular or neighborhood orderings without a commensurate change in a coarse global statistic.

This creates an identifiability problem. A geometry descriptor may move while a scorer's ordering is invariant, or a scorer may begin so close to perfect separation that little response remains observable. Before interpreting a geometry–ranking relationship, an intervention study should establish four conditions: preservation of task utility, successful manipulation of the named property, visibility of that manipulation to the scorer, and sufficient ranking headroom. Geometry-matched comparisons must additionally distinguish the matched property from the intervention history that produced it.

We study these conditions within one frozen dermatology foundation-model backbone. The study does not ask whether geometry is universally useful. It asks when a named geometric property is empirically admissible as a predictor of a named scorer's OOD ranking, and separates the question of whether a scorer can register a geometric change from the question of whether that change was successfully induced and empirically observed. AUROC is treated as a directed ranking measure [17] under a fixed higher-score-is-more-OOD convention; orientation is not reversed after inspecting domain labels.

The three research questions are:

- **RQ1 (scorer admissibility):** Which of the four evaluated distance-based scorers are, under the accepted geometry-manipulation operator, structurally capable of registering the targeted geometric change, and to what quantitative extent?
- **RQ2 (manipulability and preservation):** Can targeted geometric properties be systematically manipulated within a fixed frozen backbone while preserving in-distribution task utility?
- **RQ3 (empirical response conditional on admissibility and manipulation):** Conditional on scorer admissibility and successful geometry manipulation, do observed changes in distance-based OOD ranking support a scorer-visible geometry effect rather than intervention history or limited ranking susceptibility?

RQ1 is answered analytically in Section 3, prior to and independent of any dose-response data; RQ2 and RQ3 are answered empirically in Section 4, with RQ3 read conditional on both preceding answers.

The contributions are three.

Scorer-relevant invariance formalization (RQ1). We formalize scorer-relative transformation groups and give the resulting invariance guarantees a quantitative complement: bounded violation under ridge regularization and anisotropic mapping, connected to possible AUROC change through the ranking-margin distribution. A geometry metric earns interpretation as a predictor of ranking only once it is shown to be visible to the scorer in question rather than assumed to be; we supply that step, quantitatively, for three of the four evaluated scorers and qualitatively for the fourth, prior to and independent of any dose-response data.

Gated within-backbone intervention protocol (RQ2). We hold backbone, extraction pipeline, feature layer, splits, evaluation protocol, and scorer definitions fixed, and intervene on the representation output alone, behind an implementation-integrity gate and an explicit manipulation check. Cross-backbone studies vary architecture, pretraining corpus, objective, dimension, and feature layer together, so an observed geometry–performance covariation there may

reflect backbone provenance rather than geometry itself; holding provenance fixed removes that confound. The protocol was internally prespecified and checksum-verified before the canonical runs; it was not externally registered.

Empirical boundary evidence in a near-ceiling regime (RQ3). The contribution is not that geometry fails to predict distance-based OOD ranking. It is a characterization, conditional on RQ1's admissibility classification and RQ2's manipulation evidence, of where the question can be asked under this backbone and domain pair: in-distribution task utility remains above the prespecified admissibility threshold and the targeted geometry moves, so a trivial utility-collapse explanation is excluded; the prespecified metrics nonetheless do not account for ranking in the evaluated configuration; and reference-arm AUROC above 0.99 for two of the four estimators, and above 0.98 for a third, limits empirical identification. Where an admissible scorer's response is flat, this study reports that flatness as consistent with — but not distinguished from — a governing invariance bound or limited ranking susceptibility.

2 Related Work

2.1 Distance-Based OOD Ranking

Mahalanobis OOD detection fits class means and a tied covariance matrix [1]. Deep nearest-neighbor detection replaces the shared-Gaussian assumption with non-parametric proximity [2]. Locally adapted likelihood ratios further show that ordinary neighborhood scores may ignore locally relevant dimensions in pretrained spaces [3]. These works establish that geometry matters through the scorer that reads it. Our question is complementary: when the scorer is fixed, does a controlled change in a scorer-visible property predict a ranking change?

2.2 Representation Geometry and Intervention

The closest motivation is the reported cross-backbone association of intrinsic dimensionality, within-class spectral properties, normalization, and Mahalanobis OOD performance [4]. LID is estimated from local neighbor-distance ratios [5]. Spectrum modulation and local-dimensionality regularization show that these properties can be manipulated during representation learning [6,7]. Whitening is not unique: transforms can share a covariance target while differing in rotational or cross-correlation structure [8]. Spectral tempering provides a dose-controlled eigenvalue transformation [9]. Residual adapters modify frozen representations efficiently [10]; we optimize them with AdamW [11].

2.3 Dermatology Foundation Models and Domain Shift

PanDerm is a multimodal dermatology foundation model pretrained on more than two million images across clinical sources and modalities [12]. ISIC 2019 contains dermoscopic images assembled from constituent sources including HAM10000, ISIC challenge data, and BCN20000 [13–15]. PAD-UFES-20 instead contains clinical smartphone images [16]. The ISIC–PAD comparison is therefore cross-source and cross-modality. PAD-UFES-20 participated in development and operator calibration and is not described as an unseen OOD domain.

3 Formalization and Methods: Answering RQ1

3.1 Problem Setup

Let $f : \mathcal{X} \to \mathbb{R}^d$ be a frozen encoder, $z = f(x)$, D_{ID} the reference set, and T_θ an intervention. For scorer j, $s_j(T_\theta z; T_\theta D_{\text{ID}})$ is oriented so that larger values indicate greater OOD likelihood. Directed ranking performance is

$$A_j(\theta, D) = \Pr[s_j(T_\theta z_u) > s_j(T_\theta z_{id})]\,. \tag{1}$$

A geometry descriptor $g_k(T_\theta D_{\text{ID}})$ summarizes a named property. The empirical target is whether paired Δg_k predicts paired ΔA_j, conditional on preservation, successful manipulation, scorer visibility, and headroom.

3.2 Scorer-Relative Invariance and Violation

Definition 1 (rank invariance). A scorer s is rank-invariant to T on D if, for every evaluation pair z_a, z_b,

$$s(Tz_a; TD) < s(Tz_b; TD) \iff s(z_a; D) < s(z_b; D). \tag{2}$$

Rank invariance implies directed AUROC invariance because AUROC is the fraction of correctly ordered ID–OOD pairs.

Theorem 1 (affine Mahalanobis invariance). Let $T(z) = Az + b$ with invertible A. If class means and the shared covariance are consistently refitted after transformation and no fixed ridge is added, then for every class c,

$$(Tz - \mu_c')^\top (\Sigma')^{-1} (Tz - \mu_c') = (z - \mu_c)^\top \Sigma^{-1} (z - \mu_c). \tag{3}$$

Proof. Refitting gives $\mu_c' = A\mu_c + b$ and $\Sigma' = A\Sigma A^\top$. Therefore $Tz - \mu_c' = A(z - \mu_c)$ and $(\Sigma')^{-1} = A^{-\top}\Sigma^{-1}A^{-1}$. Substitution cancels $A^\top A^{-\top}$ and $A^{-1}A$, yielding the stated equality. Taking the minimum over classes preserves the complete score and hence its ranking. □

The exact result follows affine covariance equivariance. Its use here is scorer-specific: it identifies representation changes that a refitted OOD functional cannot observe. A fixed isotropic ridge breaks this equivariance in a quantifiable way.

Theorem 2 (ridge-induced Mahalanobis violation). Let $y = A(z - \mu_c)$, $C = A\Sigma A^\top$, $B = C + \varepsilon I$, and $D = C + \varepsilon AA^\top$, where A is invertible, $\Sigma \succ 0$, and $\varepsilon > 0$. Define $q_B = y^\top B^{-1} y$ and $q_D = y^\top D^{-1} y$. Then

$$|q_B - q_D| \le \frac{\varepsilon \|y\|^2 \|I - AA^\top\|_2}{[\lambda_{\min}(C) + \varepsilon][\lambda_{\min}(C) + \varepsilon\sigma_{\min}^2(A)]}. \tag{4}$$

Moreover, $q_D = (z - \mu_c)^\top (\Sigma + \varepsilon I)^{-1} (z - \mu_c)$.

Proof. The resolvent identity gives $B^{-1} - D^{-1} = B^{-1}(D - B)D^{-1}$, where $D - B = \varepsilon(AA^\top - I)$. Submultiplicativity gives the numerator in (4). The denominator follows from lower bounds on $\lambda_{\min}(B)$ and $\lambda_{\min}(D)$. Finally, $D = A(\Sigma + \varepsilon I)A^\top$, so affine cancellation gives the last equality. □

The bound vanishes when $\varepsilon = 0$ or $AA^\top = I$, and increases with departure from orthogonality and poor conditioning. It bounds score deviation, not AUROC change by itself.

Lemma 1 (cosine distortion). Let A be invertible and

$$\eta_A = \frac{\kappa_2(A)^2 - 1}{\kappa_2(A)^2 + 1} < 1. \tag{5}$$

For nonzero x, y,

$$|\cos(Ax, Ay) - \cos(x, y)| \le \min\left\{2, \frac{2\eta_A}{1 - \eta_A}\right\}. \tag{6}$$

Proof. Let $\kappa = \kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ and $m = (\sigma_{\max}^2 + \sigma_{\min}^2)/2$, the midpoint of the extreme eigenvalues of $A^\top A$. Write $A^\top A = m(I + E)$. The eigenvalues of E are $\sigma_i^2/m - 1$ for the singular values σ_i of A, each lying in

$$\left[\frac{\sigma_{\min}^2}{m} - 1,\ \frac{\sigma_{\max}^2}{m} - 1\right] = \left[-\frac{\sigma_{\max}^2 - \sigma_{\min}^2}{\sigma_{\max}^2 + \sigma_{\min}^2},\ \frac{\sigma_{\max}^2 - \sigma_{\min}^2}{\sigma_{\max}^2 + \sigma_{\min}^2}\right] = [-\eta_A, \eta_A],$$

using $\sigma_{\max}^2 = \kappa^2 \sigma_{\min}^2$. Hence E is symmetric with $\|E\|_2 \le \eta_A$.

For nonzero x, y, write $\hat{x} = x/\|x\|$, $\hat{y} = y/\|y\|$, $c_0 = \hat{x}^\top \hat{y} = \cos(x, y)$, and set

$$a = \hat{x}^\top E \hat{x}, \qquad b = \hat{y}^\top E \hat{y}, \qquad g = \hat{x}^\top E \hat{y}. \tag{7}$$

Since E is symmetric with $\|E\|_2 \le \eta_A$, the Rayleigh quotient bounds give $a, b \in [-\eta_A, \eta_A]$; by Cauchy–Schwarz, $|g| = |\hat{x}^\top E \hat{y}| \le \|E\|_2 \|\hat{x}\| \|\hat{y}\| \le \eta_A$.

From $x^\top A^\top A y = m\, x^\top (I + E) y$ and $x = \|x\|\hat{x}$, $y = \|y\|\hat{y}$,

$$x^\top A^\top A y = m\|x\|\|y\|\,(c_0 + g), \quad \|Ax\|^2 = m\|x\|^2(1 + a), \quad \|Ay\|^2 = m\|y\|^2(1 + b).$$

The common factor $m\|x\|\|y\|$ cancels between numerator and denominator of $\cos(Ax, Ay) = x^\top A^\top A y/(\|Ax\|\|Ay\|)$, giving the exact identity

$$\cos(Ax, Ay) = \frac{c_0 + g}{\sqrt{(1 + a)(1 + b)}}. \tag{8}$$

Because $a, b \in [-\eta_A, \eta_A]$, the product $(1 + a)(1 + b)$ lies in $[(1 - \eta_A)^2, (1 + \eta_A)^2]$, so for $\eta_A < 1$,

$$\sqrt{(1 + a)(1 + b)} \in [1 - \eta_A,\ 1 + \eta_A], \qquad \left|1 - \sqrt{(1 + a)(1 + b)}\right| \le \eta_A.$$

Rearranging (8),

$$\cos(Ax, Ay) - c_0 = \frac{c_0\left(1 - \sqrt{(1 + a)(1 + b)}\right) + g}{\sqrt{(1 + a)(1 + b)}}.$$

Using $|c_0| \le 1$, $|g| \le \eta_A$, the bound above on $|1 - \sqrt{(1 + a)(1 + b)}|$, and the denominator lower bound $\sqrt{(1 + a)(1 + b)} \ge 1 - \eta_A$,

$$|\cos(Ax, Ay) - \cos(x, y)| \le \frac{\eta_A + \eta_A}{1 - \eta_A} = \frac{2\eta_A}{1 - \eta_A}.$$

Since $\cos(Ax, Ay), \cos(x, y) \in [-1, 1]$, their difference is also trivially at most 2, giving the stated bound $\min\{2,\ 2\eta_A/(1 - \eta_A)\}$. □

For spectral tempering,

$$\kappa_2(A_\gamma) = \kappa_2(\Sigma)^{|\gamma|/2}. \tag{9}$$

Thus small $|\gamma|$ yields a near-identity angular perturbation when the reference spectrum is not

extreme. This conservative bound does not constitute a KDE bound; KDE also depends on kernel, bandwidth, and local reference mass.

3.3 Score Distortion, Ranking, and Spectral Attenuation

Let $M = s(z_u) - s(z_{id})$ be a baseline ID–OOD margin, and suppose $|\widetilde{s}(z) - s(z)| \le \delta$ on both evaluated populations.

Proposition 1 (score perturbation–ranking reversal). Let $\widetilde{M} = \widetilde{s}(z_u) - \widetilde{s}(z_{id})$. Then $|\widetilde{M} - M| \le 2\delta$. Consequently, a pair can change its directed ordering only if $|M| \le 2\delta$.

Proof. By the triangle inequality,

$$|\widetilde{M} - M| \le |\widetilde{s}(z_u) - s(z_u)| + |\widetilde{s}(z_{id}) - s(z_{id})| \le 2\delta. \tag{10}$$

A margin outside $[-2\delta, 2\delta]$ therefore cannot cross zero. □

Corollary 1 (AUROC perturbation through local margin mass). Under the conditions of Proposition 1,

$$|\operatorname{AUROC}(\widetilde{s}) - \operatorname{AUROC}(s)| \le \Pr(|M| \le 2\delta). \tag{11}$$

Proof. Directed AUROC is the probability of a positive ID–OOD margin. Proposition 1 shows that only pairs in the stated margin band can change their contribution to this probability. □

We call $S_s(\delta) = \Pr(|M| \le 2\delta)$ the *local ranking susceptibility*. Unlike global headroom $H_s = 1 - \operatorname{AUROC}(s)$, it measures the mass that a perturbation of a specified size can reorder. Near-ceiling AUROC alone does not imply small $S_s(\delta)$: correctly ordered pairs may still lie close to zero. Equation (11) is the bridge from an invariance-violation bound to an AUROC bound; evaluating it requires retained image-level scores.

Proposition 2 (Mahalanobis spectral attenuation). Let $\Sigma = U \operatorname{diag}(\lambda_1, \ldots, \lambda_d) U^\top \succ 0$, let $h = \sum_i c_i u_i$ be a displacement, and use ridge $\varepsilon \ge 0$. Its squared Mahalanobis contribution is

$$q_\varepsilon(h) = h^\top (\Sigma + \varepsilon I)^{-1} h = \sum_{i=1}^{d} \frac{c_i^2}{\lambda_i + \varepsilon}. \tag{12}$$

If h lies in an eigenspace with $\lambda_i \ge L$, then $q_\varepsilon(h) \le \|h\|^2/(L+\varepsilon)$. If it lies in an eigenspace with $\lambda_i \le \ell$, then $q_\varepsilon(h) \ge \|h\|^2/(\ell+\varepsilon)$.

Proof. Substitution of the orthonormal eigendecomposition gives (12); bounding each reciprocal eigenvalue on the stated spectral subspace gives the two inequalities. □

Thus equal-Euclidean-norm perturbations aligned with high-variance directions are attenuated relative to perturbations aligned with low-variance directions. This is a scorer-specific spectral mechanism and does not imply recoverability by another learned readout.

Cosine and Mahalanobis now have quantitative violation bounds; k-nearest-neighbour scoring, the remaining scorer with an exact invariance class under similarities (Section 3.4), has so far only that qualitative statement. Because the k-NN score is an order statistic over the reference set rather than a fixed point's distance, boundedness of the underlying linear map does not immediately imply boundedness of the score: the reference point realizing the k-th smallest distance can itself change after transformation. The next result shows the same singular-value bracket nonetheless holds for the order statistic, and yields a ranking-preservation guarantee directly comparable to Proposition 1.

Proposition 3 (k-NN score bracketing and ratio-margin ranking preservation). Let A be invertible with singular values $\sigma_{\min}(A) \le \sigma_{\max}(A)$ and $\kappa = \kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$.

Let $D = \{p_1, \ldots, p_n\} \subset \mathbb{R}^d$ be a fixed reference set, z a query point, and $s(z) = d_{(k)}$ the distance from z to its k-th nearest neighbour in D, where $d_{(1)} \leq \cdots \leq d_{(n)}$ are the sorted distances $\{\|z - p_i\|\}$. Let $\widetilde{s}(z)$ denote the same score computed on the transformed sets $\{Az_i\}$, Az. Then:

(a) Score bracketing.

$$\sigma_{\min}(A)\, s(z) \;\leq\; \widetilde{s}(z) \;\leq\; \sigma_{\max}(A)\, s(z). \tag{13}$$

(b) Ratio-margin ranking preservation. For query points z_u, z_{id} with $s(z_{id}) > 0$, if $s(z_u)/s(z_{id}) > \kappa_2(A)$, then $\widetilde{s}(z_u) > \widetilde{s}(z_{id})$: the pair's directed order cannot flip under A.

Proof. (a) Every point-to-point distance obeys $\sigma_{\min}(A)\|x\| \leq \|Ax\| \leq \sigma_{\max}(A)\|x\|$, applied here to $x = z - p_i$ for each i; this pointwise bound holds regardless of i's rank, so it does not by itself bracket an order statistic. Let $p_{(1)}, \ldots, p_{(n)}$ be D relabelled by increasing original distance to z, so $d_{(i)} = \|z - p_{(i)}\|$.

For the upper bound, the k points $p_{(1)}, \ldots, p_{(k)}$ all satisfy $d_{(i)} \leq d_{(k)}$ for $i \leq k$, so their transformed distances satisfy $\|A(z - p_{(i)})\| \leq \sigma_{\max}(A)\, d_{(i)} \leq \sigma_{\max}(A)\, d_{(k)}$ for all $i \leq k$. At least k transformed distances are therefore no larger than $\sigma_{\max}(A)\, d_{(k)}$, so the k-th smallest transformed distance satisfies $\widetilde{s}(z) \leq \sigma_{\max}(A)\, d_{(k)} = \sigma_{\max}(A)\, s(z)$.

For the lower bound, the $n - k + 1$ points $p_{(k)}, \ldots, p_{(n)}$ all satisfy $d_{(i)} \geq d_{(k)}$, so their transformed distances satisfy $\|A(z - p_{(i)})\| \geq \sigma_{\min}(A)\, d_{(i)} \geq \sigma_{\min}(A)\, d_{(k)}$. Hence at most $k - 1$ transformed distances can fall below $\sigma_{\min}(A)\, d_{(k)}$, so the k-th smallest transformed distance satisfies $\widetilde{s}(z) \geq \sigma_{\min}(A)\, d_{(k)} = \sigma_{\min}(A)\, s(z)$. This proves (13).

(b) By (13) applied to both queries, $\widetilde{s}(z_u) \geq \sigma_{\min}(A)\, s(z_u)$ and $\widetilde{s}(z_{id}) \leq \sigma_{\max}(A)\, s(z_{id})$. If $s(z_u)/s(z_{id}) > \kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$, then

$$\widetilde{s}(z_u) \;\geq\; \sigma_{\min}(A)\, s(z_u) \;>\; \sigma_{\min}(A) \cdot \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}\, s(z_{id}) \;=\; \sigma_{\max}(A)\, s(z_{id}) \;\geq\; \widetilde{s}(z_{id}),$$

so the order is preserved. □

Corollary 2 (k-NN ranking susceptibility). Define the *ratio-margin susceptibility*

$$S_{\mathrm{kNN}}(\kappa) = \Pr\left(\max\left\{ \frac{s(z_u)}{s(z_{id})}, \frac{s(z_{id})}{s(z_u)} \right\} \leq \kappa \right). \tag{14}$$

Under Proposition 3(b),

$$\left| \operatorname{AUROC}(\widetilde{s}) - \operatorname{AUROC}(s) \right| \;\leq\; S_{\mathrm{kNN}}(\kappa_2(A)). \tag{15}$$

Proof. Only pairs failing the ratio condition in Proposition 3(b) in both directions — equivalently, pairs with $\max\{s(z_u)/s(z_{id}),\, s(z_{id})/s(z_u)\} \leq \kappa_2(A)$ — can have their directed order changed by A; the argument is otherwise identical to Corollary 1's, with the additive margin band replaced by a multiplicative ratio band appropriate to a nonnegative, scale-covariant score. □

Equation (15) is the k-NN analogue of Corollary 1 and requires only $\kappa_2(A)$, not image-level scores, to state; evaluating S_{kNN} itself still requires the paired distance ratios on the evaluated population. For the tempering operator, $\kappa_2(A_\gamma) = \kappa_2(\Sigma)^{|\gamma|/2}$ as in (9), so Proposition 3 and Corollary 2 apply directly to A_γ once a baseline value of $\kappa_2(\Sigma)$ is reported; unlike Theorem 2's bound, they do not additionally require per-point residuals y.

Table 1: RQ1 answer: scorer admissibility under the accepted operator T_γ. Bound requirement contrasts data-free quantities against per-point residuals.

Scorer	Admissibility	Governing result	Bound requires
Mahalanobis	Exact ($\varepsilon = 0$) / bounded ($\varepsilon > 0$)	Theorem 1 / Theorem 2	per-point y (Thm. 2)
Cosine	Bounded	Lemma 1	only $\kappa_2(\Sigma)$
k-NN	Bounded	Proposition 3, Corollary 2	only $\kappa_2(\Sigma)$
KDE	Undetermined	Qualitative group only (Sec. 3.4)	Kernel and bandwidth model

3.4 Transformation-Group Interpretation

Classical group-invariance theory studies inferential quantities that remain constant on transformation orbits [21]. Here the consistently refitted unregularized Mahalanobis scorer admits $GL(d) \ltimes \mathbb{R}^d$. Cosine admits orthogonal maps and positive common scaling, but not general translation or anisotropic maps; at the pair level it is also unchanged by independent positive radial rescaling. Euclidean k-NN ordering admits similarities—translation, orthogonal maps, and uniform positive scaling—whereas KDE invariance additionally depends on bandwidth and density transformation. These groups are different and may be nested for particular subgroups; their sensitivity profiles are not identical. Scorer-visible and scorer-invisible geometry are therefore structural consequences of different transformation orbits. Together, Definition 1, Theorems 1–2, Lemma 1, Proposition 1, Corollary 1, Proposition 2, Proposition 3, and Corollary 2 replace a binary invariance label by a hierarchy of exact, bounded, ranking-level, and spectral statements, covering three of the four evaluated scorers quantitatively and the fourth (KDE) through its qualitative transformation group alone.

Answering RQ1. This section constitutes the answer to RQ1, established analytically rather than by dose-response data, and holding for the accepted operator T_γ (an exact instance of the affine class of Theorem 1 for any fixed γ, since U, $\{\lambda_i\}$, and μ are fixed once from the training covariance). Mahalanobis is exactly invisible to T_γ at zero ridge (Theorem 1) and boundedly visible under the ridge actually used in the fitted scorer (Theorem 2, vanishing as $\gamma \to 0$). Cosine and k-NN are both boundedly visible, with violation governed by the same quantity, $\kappa_2(A_\gamma) = \kappa_2(\Sigma)^{|\gamma|/2}$ (Lemma 1; Proposition 3 and Corollary 2). KDE's admissibility is not established by this formalization and is treated as undetermined rather than assumed either visible or invisible. Table 1 summarizes this classification; Section 4 reports the manipulation and empirical evidence that RQ2 and RQ3 condition on it.

3.5 Backbone, Populations, and Intervention

The frozen PanDerm ViT-L/16 encoder produces 1024-dimensional embeddings. ISIC 2019 contributes 25,331 images split with a fixed seed into training (n = 16,211), validation (n = 4,053), and held-out evaluation (n = 5,067). PAD-UFES-20 contributes 2,298 stress-test images mapped into the same eight-class label space. Embeddings are extracted once under fixed preprocessing and stored with checksums.

For seed $r \in \{42, 52, 62, 72, 82\}$, a zero-initialized residual bottleneck produces displacement $\Delta z_r = W_2\,\mathrm{GELU}(W_1 z)$. Post-training interpolation defines

$$T_{r,\alpha}(z) = z + \alpha \Delta z_r, \quad \alpha \in \{0, .25, .50, .75, 1\}. \tag{16}$$

All doses from a seed share one fitted adapter, preserving pairing. A Conventional arm uses the same architecture, schedule, budget, and seeds without the geometry-targeting term.

Gate 0 requires in-distribution eight-class balanced accuracy above chance by a fixed margin ($> 0.125 + 0.10 = 0.225$). Gate 1 requires at least one prespecified target—LID or within-class spectral-decay slope—to show reproducible dose dependence across seeds. Gate 0 is a threshold-based utility-admissibility check rather than a non-inferiority or equivalence test.

3.6 Geometry Operators and Scorers

The initial radial normalization operator changed condition number strongly but left LID and spectral slope nearly unchanged, and therefore serves as a negative manipulation check. The accepted operator uses the eigendecomposition $\Sigma = U\,\mathrm{diag}(\lambda_i)U^\top$ of the ID training covariance:

$$T_\gamma(z) = \mu + U\,\mathrm{diag}(\lambda_i^{-\gamma/2})U^\top(z-\mu), \quad \gamma \in [0,1]. \tag{17}$$

Its transformed covariance is $U\,\mathrm{diag}(\lambda_i^{1-\gamma})U^\top$, so in exact arithmetic

$$\kappa(\Sigma') = \kappa(\Sigma)^{1-\gamma}. \tag{18}$$

LID provides a complementary scale check. Under one common multiplier $c > 0$, every neighbor distance satisfies $r_j(cz) = cr_j(z)$, so every ratio r_j/r_k and the usual maximum-likelihood LID estimate are unchanged. A pointwise radial operator need not preserve these ratios, so the small LID movement under normalization is an empirical negative manipulation check rather than an exact invariance result.

Four scorers are refit on each transformed ID training population: class-conditional Mahalanobis distance with ridge 10^{-5}, cosine-to-centroid, k-NN with $k = 10$, and class-conditional KDE. Higher scores are fixed to mean more OOD. Baseline headroom is $H_j = 1 - A_{j,\mathrm{ref}}$. When image-level scores are available, the stronger diagnostic is $S_j(\delta)$ from Corollary 1, with δ supplied by a scorer-specific analytical or empirical score-distortion bound.

3.7 RQ3 Estimands

The paired configuration-level estimands are

$$\Delta g_{r,\alpha} = g_{r,\alpha} - g_{r,0}, \qquad \Delta A_{j,r,\alpha} = A_{j,r,\alpha} - A_{j,r,0}. \tag{19}$$

The available Kendall coefficients across 30 configurations are retained only as exploratory summaries because configurations sharing a seed are dependent. Mixed-effects interactions group by seed [18]; Holm correction controls the metric–scorer comparison family [19]. Hierarchical intervals require image-level score vectors [20]; aggregate AUROC means are not treated as image-level observations.

RQ3 uses three paired configurations: $A = (\alpha = 1, \gamma = 0)$; $B = (\alpha = .25, \gamma = 0)$; and $C = (\alpha = 1, \gamma = .01)$. Configuration C matches B on spectral slope and A on adapter dose/intervention history. It does not match B on LID. Paired $C - B$ and $C - A$ contrasts are tested for four scorers with Holm correction across eight comparisons. For Mahalanobis, $A - C$ is a near-invariance consistency check; it is not treated as a powered geometry-effect test.

Table 2: Manipulation and threshold-based utility-admissibility evidence.

Operator	Geometry outcome	ID task	RQ2 use
Adapter α	At least one target dose-dependent	Gate 0 pass	Positive
Norm.	ΔLID ≈ 0; slope $< 10^{-6}$	Not decisive	Negative
Tempering	Slope and LID change across γ	Gate 0 pass	Positive
$\gamma = .01$	Slope matched; LID differs by .60	Gate 0 pass	Slope only

4 Empirical Results: RQ2 and RQ3

RQ1 was answered analytically in Section 3 and does not depend on the data reported below: Table 1 classifies each scorer's admissibility under the accepted operator T_γ prior to, and independent of, any dose-response evidence. This section reports the empirical evidence for RQ2 (manipulation and preservation) and RQ3 (empirical response, read conditional on RQ1's admissibility classification and RQ2's manipulation evidence).

4.1 RQ2: Manipulation and Preservation

All Intervention and Conventional checkpoints satisfied the threshold-based Gate 0 at every dose, and the Intervention arm passed Gate 1 across all five seeds. Gate 0 is a weak instrument on its own: it asks only for balanced accuracy above .225 against a chance rate of .125, so a model degrading from .75 to .30 would pass it, and observed utility runs .5351 to .5868.

Because the two arms are run at matched doses and matched seeds, the question admits a paired answer. Across the twenty matched pairs, the mean utility difference is -0.0000 with a seed-clustered bootstrap interval of $[-.0025, +.0029]$ and a one-sided 95% lower bound of $-.0022$; per-dose means range from $-.0010$ to $+.0007$. Non-inferiority therefore holds for any margin $\delta_U > .0022$, roughly a fifth of a percentage point of balanced accuracy. We do not select δ_U here: which degradation is clinically tolerable is a prior decision, and the data cannot supply it. What the paired analysis establishes is that the intervention has no measurable utility cost at the resolution this design affords, which is a stronger statement than clearing Gate 0 and the one the word "preservation" requires. The normalization operator changed condition number by approximately -2.7×10^5 but left LID effectively unchanged (mean -4×10^{-4}, SD 1.8×10^{-3}) and changed spectral slope by less than 10^{-6} (SD 10^{-6}). It was excluded from geometry-matched interpretation.

Spectral slope changed reproducibly and near-monotonically by -1.5, -2.7, -3.6, and -3.5×10^{-3} at $\gamma = .25, .50, .75$, and 1.00. LID changed reproducibly but non-monotonically, increasing by approximately $+2.7$ to $+5.8$ at intermediate exponents before decreasing by approximately -1.5 to -4.4 at full whitening. At $\gamma = .01$, slope was 6.746×10^{-3} against a target of 6.751×10^{-3}, while LID remained 0.60 below the target checkpoint. The manipulation component of RQ2 is supported within the observed trajectories, all configurations meet the threshold-based utility gate, and the paired analysis above establishes non-inferiority for any margin above .0022. Arbitrary or joint geometric control is not established.

4.2 RQ3: Empirical Response Conditional on Admissibility and Manipulation

This section answers RQ3 using both the pooled configuration-level analysis and the geometry-matched paired contrast, conditional on RQ1's admissibility classification (Table 1) and RQ2's manipulation evidence.

Table 3: Exploratory configuration-level Kendall τ. Configurations sharing a seed are dependent.

Metric	Mahal.	Cosine	k-NN	KDE
LID	$-.24$	.01	.19	.10
Spectral slope	.22	.16	.12	.05
Condition number	.56	.58	.42	.55
Nuisance variance	.32	.34	.33	.15

Table 4: Mean directed AUROC over five paired seeds.

Scorer	$A : \alpha = 1$	$B : \alpha = .25$	$C : \alpha = 1, \gamma = .01$
Mahalanobis	.9984	.9984	.9984
Cosine	.9498	.9750	.9523
k-NN	.9928	.9973	.9930
KDE	.9967	.9986	.9968

Pooled configuration-level associations. Neither target metric had a Holm-corrected association with any scorer in the exploratory pooled analysis (Table 3). The largest LID and slope associations were both with Mahalanobis ($\tau = -.24$, uncorrected $p = .065$; $\tau = .22$, $p = .088$). Condition number was associated with all scorers in the pooled table, but no association survived whole-seed permutation sensitivity after correction. With $5! = 120$ permutations, that sensitivity analysis has coarse resolution and is read as unresolved.

Condition number's pooled association, uniform across all four scorers regardless of each scorer's admissibility class in Table 1, should not be read as support for a geometry-mediated ranking effect. RQ1 establishes that a consistently refitted Mahalanobis score is exactly insensitive to the isotropic component of a transformation (Theorem 1) and only boundedly sensitive under the ridge actually used (Theorem 2); the manipulation check in RQ2 independently shows that the normalization operator moves condition number by approximately -2.7×10^5 while leaving both prespecified shape metrics (LID, slope) essentially unchanged. Condition number therefore behaves as a scale-sensitive descriptor rather than a measure of the spectral shape the prespecified metrics target, and an association uniform across scorers with different admissibility classes is more consistent with a shared scale confound than with a scorer-visible geometric mechanism.

Three of 16 intervention-type-by-geometry interactions were Holm-significant: LID with Mahalanobis ($p = 2.9 \times 10^{-6}$), LID with KDE ($p = 2.6 \times 10^{-3}$), and condition number with Mahalanobis ($p = 3.9 \times 10^{-5}$). All slope interactions were non-significant. The significant coefficients were small and mechanism-dependent, which is inconsistent with treating geometry as a mechanism-independent predictor. Spectral slope is not supported in this stress-test regime; LID remains unresolved because it is not independently matched while holding slope and intervention history apart. The pooled coefficients do not by themselves answer the paired RQ3 estimand.

Geometry-matched paired contrast. Table 4 reports the A/B/C comparison. Mahalanobis and k-NN show little separation. For cosine and KDE, C is descriptively closer to history-matched A than to slope-matched B: $C - A$ differences are .0018 and .0046, whereas $C - B$ differences are $-.0229$ and $-.1056$. None of the eight paired comparisons reaches Holm-corrected significance; the smallest corrected p is .111 and all paired Wilcoxon p-values are at least .063. The pattern is therefore descriptive, not proof that ranking follows intervention history.

Table 5: Global headroom and A/B/C response range on the calibration pair. Local susceptibility is reported separately in Table 6.

Scorer	Ref.	H_j	Range	Class
Mahalanobis	.9984	.0016	.0000	Near ceiling
k-NN	.9973	.0027	.0046	Near ceiling
Cosine	.9750	.0250	.0251	Limited
KDE	.9986	.0014	.0019	Near ceiling

Table 6: Local ranking susceptibility from reconstructed image-level scores. $S(\delta) = \Pr(|M| \leq 2\delta)$ at the observed $\alpha : 0 \to 1$ score distortion. Only the $\delta_{\max}$ column is a valid bound; the median column is descriptive and its value is the ordering across scorers, not whether it exceeds the observation.

Scorer	δ_{med}	$S(\delta_{\text{med}})$	$\lvert\Delta A\rvert$	$\delta_{\max}$	$S(\delta_{\max})$
Mahalanobis	7.44	.000065	.000160	1.08×10^4	.690
Cosine	.0098	.0478	.0337	.301	1.000
k-NN	1.09	.0073	.0051	2.90	.832
KDE	101.3	.0186	.0021	7.72×10^3	1.000

Image-level scores were regenerated from the frozen embedding artifact and the retained adapters, reproducing every stored aggregate AUROC to six decimals, so the local susceptibility that aggregate means cannot express is available directly (Table 6). Two readings follow. First, the bound of Corollary 1 is valid only at $\delta_{\max}$, and there it is uninformative: $S = 1.000$ for cosine and KDE, .832 for k-NN, .690 for Mahalanobis. Second, at a common descriptive δ the scorers separate by three orders of magnitude, with Mahalanobis at 6.5×10^{-5} against cosine's 4.8×10^{-2} — a factor of 735 — even though Mahalanobis's score displacement is the larger of the two (median 7.44, 95th percentile 2.5×10^3). Mahalanobis's flat profile is therefore ranking insensitivity, not a ceiling artifact. This distinction is what the pooled aggregates could not make.

External validation. Three additional evaluation pairs were added under a protocol fixed before any external image was embedded: Derm7pt (Interactive Atlas of Dermoscopy) in both its dermoscopic and clinical arms, and PH2. Neither source contributes to ISIC 2019, and a near-duplicate screen in encoder space flagged no image in any set against the 21,278 frozen ISIC embeddings, at a threshold calibrated as the 99.9th percentile of within-ISIC nearest-neighbour similarity. The Derm7pt arms photograph the same 983 lesions under two modalities, which separates the modality gap from the site gap in a way the calibration pair cannot.

The same-modality pair supplies the headroom the calibration pair lacks: baseline cosine AUROC falls from .9805 to .7805, and to .6322 on PH2 (Table 7). Two results follow, and they answer different questions.

The direct contrast replicates. $C - A$, which holds adapter dose fixed and varies only the tempering exponent, is positive and excludes zero for cosine on all three external pairs, with the clinical pair landing within 2×10^{-4} of the calibration estimate. k-NN holds on both Derm7pt arms. Mahalanobis remains flat everywhere, as Theorem 2 and Table 6 jointly predict. Changing representation geometry changes distance-based OOD ranking, reproducibly and across domains that differ in site, modality and difficulty.

The geometry-matched contrast does not. Holding G fixed and varying intervention identity yields no interval excluding zero in any of the twelve external pair–scorer combinations, with every negative-control permutation above $p = .2$ against $p = .006$ on the calibration pair. Bal-

Table 7: External validation. $C - A$ holds $\alpha = 1$ and adds tempering, so the two configurations differ in geometry; the matched contrast holds geometry fixed and varies intervention identity. Intervals are hierarchical bootstrap over seeds then examples.

	PAD-UFES (calib.)	Derm7pt dermoscopic	Derm7pt clinical	PH2
Baseline AUROC (cosine)	.9805	.7805	.9340	.6322
$C - A$ cosine	+.00241	+.00361	+.00229	+.00423
excludes zero	yes	yes	yes	yes
$C - A$ k-NN	+.00022	+.00111	+.00091	+.00022
$C - A$ Mahalanobis	+.00000	+.00001	+.00000	−.00001
Matched contrasts excluding zero	2/4	0/4	0/4	0/4

ance is better on the external pairs, not worse (worst |SMD| .043 and .054 against .136), and the calibration estimate for cosine falls outside both Derm7pt intervals, so this is a contradiction rather than a failure to resolve. Three controls exclude a pipeline explanation: recomputed geometry matches the stored values to 5×10^{-5} relative and the AUROCs exactly; the same code reproduces the calibration result to 0.000×10^{0} on the estimate, both interval bounds and the pair count; and subsampling PAD-UFES to the external sample size over forty stratified draws leaves the cosine contrast at $-.00850$ excluding zero in 100% of draws, so sample size is not the explanation.

Read together the two results are the two halves of one statement. Geometry change moves ranking, reliably. Intervention identity, once geometry is held fixed, adds nothing. Within the evaluated operator family the intervention acts *through* the measured geometry rather than beyond it, and RQ3 is answered rather than left unresolved. The scope qualifier is that this holds for the two-coordinate G evaluated here; a richer geometry vector could in principle leave residual identity structure that (LID, slope) does not capture.

One asymmetry deserves emphasis rather than a footnote. PAD-UFES-20 is the only evaluation domain that participated in operator calibration, and it is the only domain on which the matched contrast is nonzero. We do not claim this establishes calibration leakage, but the ordering of the evidence is consistent with it, and it is a more specific candidate explanation than a ceiling artifact.

The equal rounded Mahalanobis AUROC for A and C is consistent with the small ridge-induced deviation allowed by Theorem 2, not evidence of exact numerical identity. The theorem does not constrain $A - B$, which differs in nonlinear adapter dose, and makes no exact prediction for the other scorers under anisotropic tempering. k-NN's near-flat profile across A, B, and C is likewise consistent with the bracket of Proposition 3 and the ratio-margin susceptibility of Corollary 2: both bounds are governed by the same $\kappa_2(A_\gamma) = \kappa_2(\Sigma)^{|\gamma|/2}$ that bounds cosine's deviation under Lemma 1, so a small violation is expected for k-NN whenever it is expected for cosine, but Table 5's near-zero headroom for k-NN means this bound cannot by itself be distinguished from a ceiling effect with the retained aggregates. Lemma 1 predicts that angular deviation contracts as $\kappa(A_\gamma) \to 1$, but the retained aggregates do not permit a numerical bound-to-observation comparison for cosine either. Minimal global headroom likewise does not establish irrelevance: Corollary 1 and Corollary 2 both require local margin or ratio-margin mass, which cannot be recovered from Table 5. With image-level scores available, Table 6 distinguishes what the aggregates could not: Mahalanobis's and k-NN's flatness is governed by their low ranking susceptibility rather than by a ceiling, and the external pairs supply the headroom needed to test the matched property where no ceiling applies.

5 Discussion and Limitations

The three RQs delineate a methodological boundary rather than a general verdict on geometry. RQ1 establishes, analytically and prior to any dose-response data, which scorers are structurally capable of registering the accepted operator's geometric change and at what quantitative cost. RQ2 establishes controlled movement of target metrics while in-distribution task utility remains above a prespecified admissibility threshold. RQ3, read conditional on both, shows why pooled associations require scorer-visibility screening before interpretation, and separates two questions that a single flat curve conflates: whether changing geometry changes ranking, and whether intervention identity matters once geometry is fixed. On external domains with headroom the first is answered affirmatively and reproducibly, the second negatively, and together they place the intervention's effect inside the measured geometry rather than beyond it.

The central lesson is an admissibility rule: a geometry–OOD claim requires successful manipulation, preservation, scorer visibility, and sufficient local ranking susceptibility. Failure of any condition changes the meaning of a null response. Exact group invariance supplies a theoretical zero; Theorem 2 and Lemma 1 replace the remaining binary judgment by scorer-specific quantitative bounds; Proposition 1 and Corollary 1 connect those bounds to ranking and AUROC; and Proposition 2 identifies a concrete spectral attenuation mechanism for Mahalanobis scoring. Global headroom remains a useful warning but is not a substitute for the pairwise margin distribution. None of these conditions supports a universal conclusion that geometry is irrelevant.

The evaluation covers one backbone, one layer, one diagnostic task, five optimization seeds, and four evaluation domain pairs, of which three are locked and same-modality or lesion-matched. Generalization beyond a single encoder is the principal remaining limitation: every statement here is, strictly, a statement about PanDerm ViT-L/16, and whether the susceptibility ordering across scorers is a property of distance-based OOD detection or of this representation cannot be settled with one backbone.

Five seeds estimate the seed-level variance component coarsely, so hierarchical intervals are narrower than a seed-only analysis and should not be read as though more seeds were available. The observed between-seed spread of the paired contrast exceeds the Hanley–McNeil sampling standard error by up to a factor of two for cosine, and minimum detectable effects here are computed from the observed spread rather than the formula for that reason. PH2 contributes 200 OOD images, which yields a minimum detectable effect of .059 for cosine; it is retained as a second point on the difficulty ladder and does not stand alone.

LID and spectral slope are correlated at $-.836$ on the calibration pair, so "after geometry is accounted for" is ambiguous there in a way it is less so on the external pairs ($-.53$ to $-.64$); matching on a two-coordinate G whose coordinates are nearly redundant is closer to matching on one. RQ3 matches slope alone at $\gamma = .01$; LID remains 0.60 below target on the calibration pair, and 0.26 when averaged over five seeds rather than read from seed 42.

The near-duplicate screen operates in encoder space because the ISIC source images were not retained. It is a weaker instrument than a perceptual hash for exact or cropped duplicates, though a stronger one for the same lesion rephotographed, and it can only fail to find evidence of overlap rather than establish its absence.

6 Conclusion

Within a frozen dermatology foundation-model backbone, targeted representation geometry can be manipulated while in-distribution task utility remains above a prespecified admissibility

threshold, and on locked external domains with headroom that manipulation moves distance-based OOD ranking reproducibly. Holding the measured geometry fixed, the intervention that produced a configuration adds nothing: no geometry-matched contrast excludes zero on any external pair, while the direct tempering contrast excludes zero for cosine on all of them. Within the evaluated operator family and the two-coordinate geometry vector used here, the intervention acts through geometry rather than beyond it.

Scorer-relative transformation groups identify exact invisibility; perturbation bounds quantify departures from it; and local margin mass, rather than global headroom alone, bounds the AUROC response that a score perturbation can produce. Reconstructed image-level scores separate ranking insensitivity from a ceiling, a distinction aggregate AUROC cannot express and one that changes what a null response means. Geometry–OOD claims require manipulation, utility admissibility, scorer visibility, and evaluation susceptibility before a ranking response can be interpreted; the last of these is the condition most easily mistaken for evidence of invariance.

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